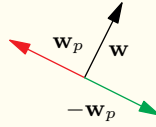


1.1.1

Gitt en vektor $\mathbf{w}$. Vi definerer $\mathbf{w}_p$ som vektoren som

- står vinkelrett på $\mathbf{w}$.
- peker til venstre med hensyn til retningen til $\mathbf{w}$.
- har lik lengde som $\mathbf{w}$.

Hvis $\mathbf{w} = [x, y]$, kan det vises at $\mathbf{w}_p = [-y, x]$, og at $-\mathbf{w}_p$ peker til høyre for $\mathbf{w}$.



Gitt vektorene $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$ og $\mathbf{v}_p = [-d, c]$. Vis at

- $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$
- $(\mathbf{u}_p)_p = -\mathbf{u}$
- $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Svar

$$\text{a) } [a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$$

$$\text{b) } (\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}.$$

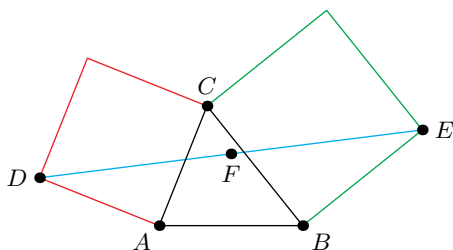
c)

$$\begin{aligned} (\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\ &= [-(b + d), a + c] \\ &= [-b, a] + [-d, c] \\ &= \mathbf{u}_p + \mathbf{v}_p \end{aligned}$$

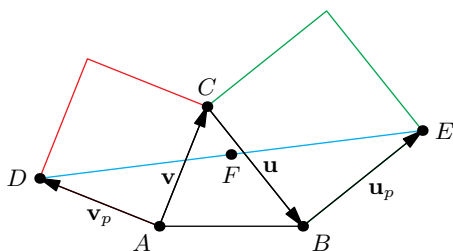
1.1.2

Den røde og den grønne firkanten er begge et kvadrat. Vi setter $\mathbf{v} = \overrightarrow{AC}$ og $\mathbf{u} = \overrightarrow{CB}$. F er midtpunktet på DE .

- Finn koordinatene til F uttrykt ved $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$ og $\mathbf{v}_p$.
- Beskriv den eventuelle forflytningen til F hvis C forskyves i en vilkårlig retning. (Det holder å se på tilfellet hvor C ligger over AB).



Svar



Vi innfører et koordinatsystem med origo i A . Da er $D = \mathbf{v}_p$. Videre er $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Videre er $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Dermed er

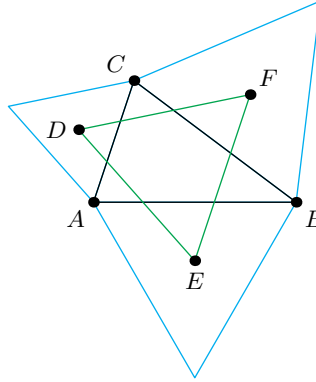
$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

$\mathbf{u} + \mathbf{v}$ og $\mathbf{u}_p + \mathbf{v}_p$ er ortogonale og har lik lengde. Så lenge $\mathbf{u} + \mathbf{v}$ ikke endres, vil altså ikke F forflyttes. En forskyvning av C medfører ingen endring av $\mathbf{u} + \mathbf{v}$, og dermed er posisjonen til F uavhengig av posisjonen til C .

1.1.3

De blå linjestykkene inngår i likesidede trekanter på utsiden av $\triangle ABC$. D , E og F er de respektive midtpunktene til de likesidede trekantene.

Vis at $\triangle DEF$ er likesidet.



.

Svar

Vi setter $\overrightarrow{AC} = 2\mathbf{v}$ og $\overrightarrow{AB} = 2\mathbf{u}$. Da er $\overrightarrow{BC} = 2\mathbf{v} - 2\mathbf{u}$. Videre har vi at

$$D = \mathbf{v} + k\mathbf{v}_p$$

$$E = \mathbf{u} - k\mathbf{u}_p$$

$$F = \mathbf{u} + \mathbf{v} - k(\mathbf{v}_p - \mathbf{u}_p)$$

hvor $k = \frac{1}{\sqrt{3}}$. Altså er

$$\overrightarrow{DF} = \mathbf{u} + k(\mathbf{u}_p - 2\mathbf{v}_p)$$

$$\overrightarrow{DE} = \mathbf{u} - \mathbf{v} - k(\mathbf{u}_p + \mathbf{v}_p)$$

$$\overrightarrow{EF} = \mathbf{v} + k(2\mathbf{u}_p - \mathbf{v}_p)$$

Da er

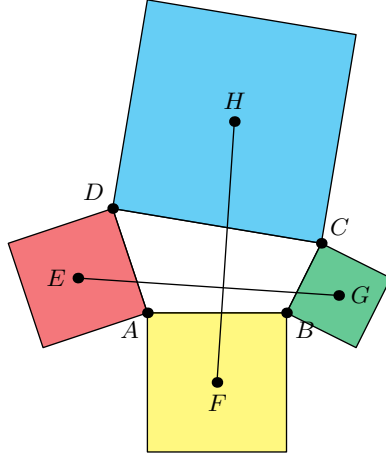
$$\begin{aligned} DF^2 &= \mathbf{u}^2 + 2k\mathbf{u}(\mathbf{u}_p - 2\mathbf{v}_p) + k^2(\mathbf{u}_p - 2\mathbf{v}_p)^2 \\ &= \mathbf{u}^2 - 4k\mathbf{u} \cdot \mathbf{v}_p + k^2(\mathbf{u}_p^2 - 4\mathbf{u}_p \cdot \mathbf{v}_p + 4\mathbf{v}_p^2) \\ &= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p \end{aligned}$$

$$\begin{aligned}
DE^2 &= (\mathbf{u} - \mathbf{v})^2 - 2k(\mathbf{u} - \mathbf{v})(\mathbf{u}_p + \mathbf{v}_p) + k^2(\mathbf{u}_p + \mathbf{v}_p)^2 \\
&= \mathbf{u}^2 - 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2 - 2k(\mathbf{u} \cdot \mathbf{v}_p - \mathbf{u}_p \cdot \mathbf{v}) + k^2(\mathbf{u}_p^2 + 2\mathbf{u}_p \cdot \mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p
\end{aligned}$$

$$\begin{aligned}
EF^2 &= \mathbf{v}^2 + 2k\mathbf{v}(2\mathbf{u}_p - \mathbf{v}_p) + k^2(2\mathbf{u}_p - \mathbf{v}_p)^2 \\
&= \mathbf{v}^2 + 4k\mathbf{u}_p \cdot \mathbf{v} + k^2(4\mathbf{u}_p^2 - 4\mathbf{u}_p \mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p
\end{aligned}$$

1.1.4

Firkantene på utsiden av $\square ABCD$ er kvadrater. E , F , G og H er de respektive midtpunktene til kvadratene. Vis at $EG = FH$ og $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Vi innfører et koordinatsystem med origo i A , og setter $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, og $\overrightarrow{BC} = 2\mathbf{w}$. Da er

$$E = -\mathbf{v} - \mathbf{v}_p$$

$$F = \mathbf{u} - \mathbf{u}_p$$

$$G = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p$$

$$H = \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

Dermed er

$$\overrightarrow{EG} = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p$$

$$\overrightarrow{FH} = \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

Dermed er

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

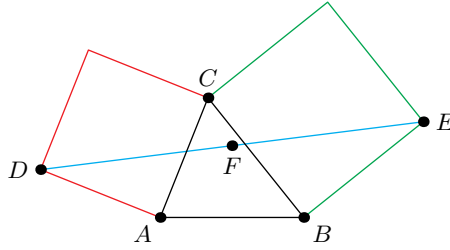
Dette betyr at $\overrightarrow{FH}$ og $\overrightarrow{EG}$ er ortogonale. Videre er

$$FH = |\overrightarrow{FH}_p| = |-\overrightarrow{EG}| = EG$$

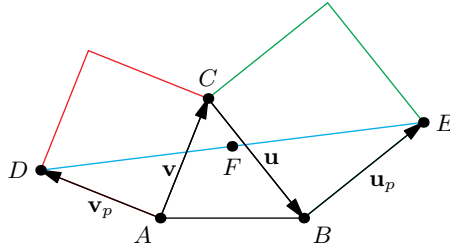
Problem: Bottema's theorem

The red and the green quadrilateral are both squares. F is the midpoint of DE . Let $\mathbf{v} = \overrightarrow{AC}$ and $\mathbf{u} = \overrightarrow{CB}$. Given a vector $\mathbf{w}$, we define $\mathbf{w}_p$ as the vector perpendicular to $\mathbf{w}$, pointing to the left with respect to the direction of $\mathbf{w}$, and with equal length as $\mathbf{w}$.

- Find the coordinates of F expressed in terms of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$, and $\mathbf{v}_p$.
- Describe the possible displacement of F if C is shifted in an arbitrary direction. (It is sufficient to consider the case where C is above AB).



Solution



We introduce a coordinate system with the origin at A . Then $D = \mathbf{v}_p$. Furthermore, $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Furthermore, $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Thus,

$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

$\mathbf{u} + \mathbf{v}$ and $\mathbf{u}_p + \mathbf{v}_p$ are orthogonal and have the same length. As long as $\mathbf{u} + \mathbf{v}$ does not change, F will not be displaced. A displacement of C does not change $\mathbf{u} + \mathbf{v}$, and thus the position of F is independent of the position of C .